Thong Pham

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EDUCATION

University of Colorado, Boulder **Boulder, CO**
Master of Science: Computer Science (**Specialization: Machine Learning, GPA 4.0 / 4.0**) *June 2024 - December 2025*

- **Relevant Courses**: Advance Data Structure, RSA, and Quantum Algorithm | Approx of Algorithm and Linear Programming | Deep Learning for Natural Language Processing | Introduction to Generative AI | Introduction to Computer Vision

University of California, Irvine **Irvine, CA**
Bachelor of Arts: Business Economics *September 2019 - December 2022*

WORK EXPERIENCE

Mercor **San Francisco, CA**
Software Engineer & Tooling *February 2025 - Present*

- Built Python tooling that runs GPT-5, Gemini, and other LLM APIs to automatically validate dataset and label quality for multiple AI labs, **increasing testing throughput by ~40% and accelerating their R&D cycles**.
- Designed and maintained data pipelines **from Snowflake into Hex**, delivering analysis dashboards that track model performance, label coverage, and **data-quality trends while cutting manual data processing time by ~25%**.
- Architected Airtable bases, views, and automations to manage large-scale data labeling workflows end-to-end, improving labeler throughput and standardising quality checks for training data.
- Partnered with ML and infrastructure engineers to harden evaluation jobs and data pipelines, **improving system reliability by ~50% across staging and production environments**.

Software Engineer LLM

- Resolved hundreds of LLM and data-related issues each week across Python, Java, and C/C++ services integrating GPT-5, Gemini, and other model APIs. Maintained **a first-pass fix rate above 99%**.
- Debugged and optimized API integration paths, rate limiting, and error handling for LLM workloads, significantly reducing incident noise and on-call load for AI lab partners.
- Ensured all code changes met strict guidelines and review standards required by top AI research labs, keeping **major post-merge defect rates below 1% and shortening review cycles**.

Kashmir World Foundation **Richmond, VA**
Machine Learning Engineer Intern *September 2024 - February 2025*

- Utilized **XGBoost**, **PyTorch**, and **OpenCV** on **large scale bioacoustic data** (over 200K audio and video clips) to achieve **91.6% classification accuracy**, demonstrating end-to-end data ingestion, hyperparameter tuning, and deployment.
- Performed **exploratory data analysis (EDA)** and **feature selection** (random forests, correlation analysis) on complex signal based datasets to enhance model interpretability for domain experts studying endangered species.
- Create interactive **Power BI** and **Tableau** dashboards to present modeling insights and trend analyses, informing **conservation researchers** and decision-makers on species behavior and habitat threats.

Stealth Startup **Irvine, CA**
Software Engineer *February 2023 – June 2024*

- Implemented LLM-based product features for finance workflows using hosted model APIs, including prompt design, API orchestration, and output post-processing in Python on transaction and account data.
- Built lightweight evaluation and regression scripts to track LLM behavior on core financial use cases (e.g., report generation, risk/exception handling explanations) and turn production bugs into repeatable test cases.

PROJECTS

Retro Kong AI (PPO) | *Python, PPO (Proximal Policy Optimization), HuggingFace* *January 2025*

- Developed an AI agent to play the 1980s arcade game "**Kong**," with barrels rolling toward the player's character. **Using Proximal Policy Optimization** in a custom environment mirroring original game physics, the agent refined its platform navigation and dodging strategies. **By tuning learning rate and clip range**, it exhibited near human reflexes over multiple training epochs, illustrating PPO's effectiveness in mastering classic arcade challenges.

TECHNICAL SKILLS

Programming Languages: Python (TensorFlow, Scikit-learn, Keras, PyTorch, Seaborn, OpenCV), R, Java, C/C++, SQL.
Tools & Frameworks: Git (CI/CD), PowerBI, Tableau, Airtable, Docker, Excel.
Data Engineering & Big Data: AWS (S3, EC2), Google Cloud (GCP), Snowflake.
Language: English (Native), Vietnamese (Native), Spanish (Intermediate).